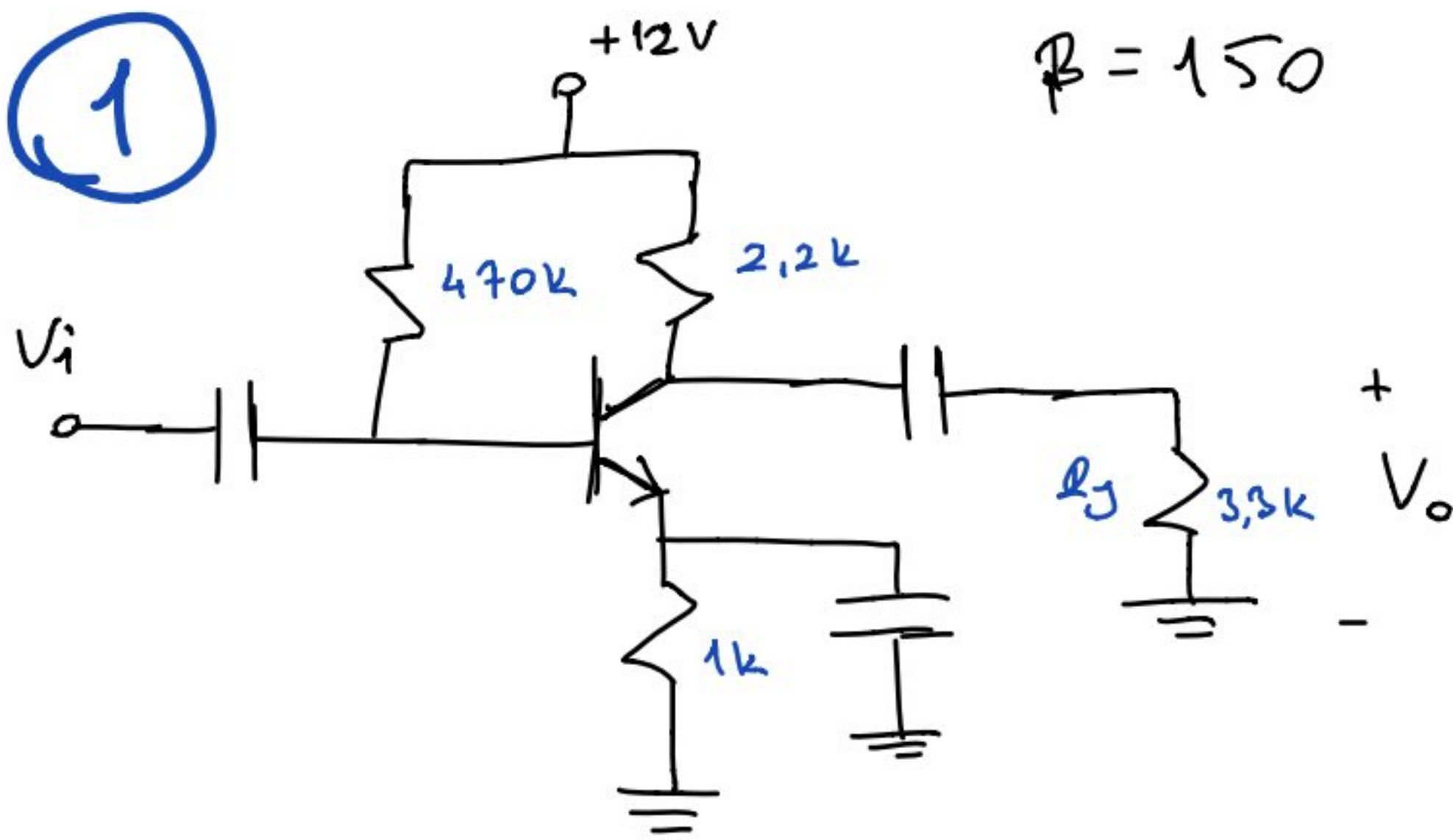


1

$$\beta = 150$$



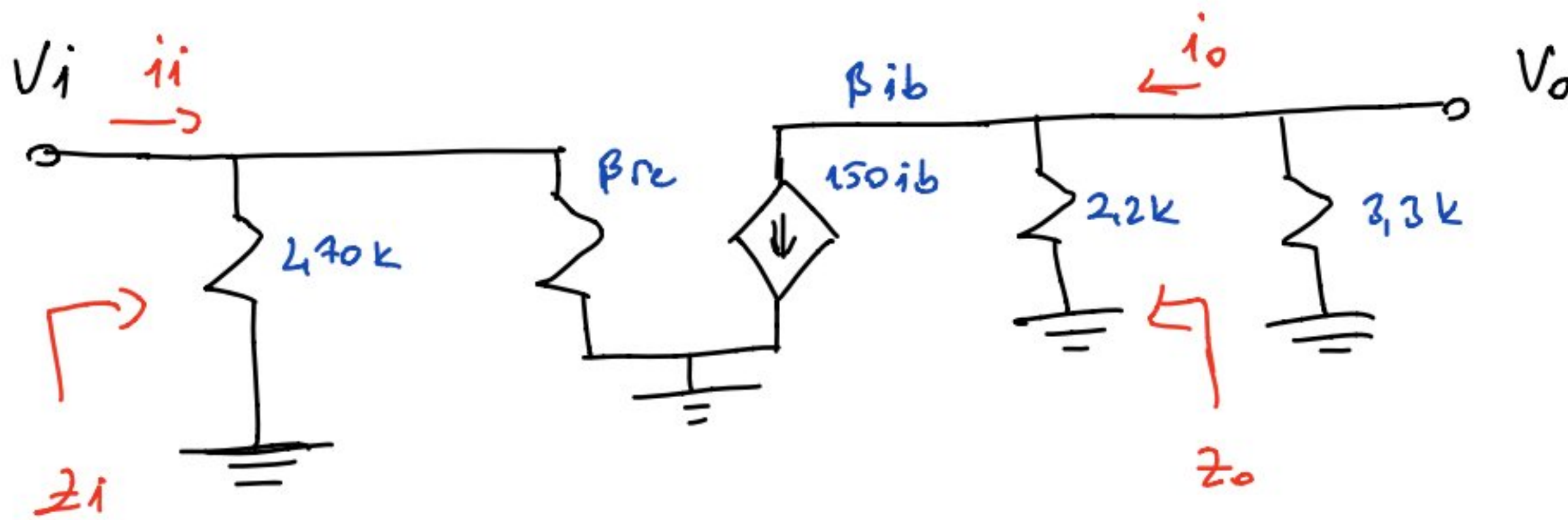
a) $r_e = ?$

b) $z_i = ?$

c) $z_o = ?$

d) $A_v = V_o / V_i = ?$

e) $A_i = i_o / i_i = ?$



a) $r_e = ?$ $r_e = \frac{26mV}{I_{E_{DC}}}$ $I_E = I_B (\beta + 1)$

$$12 = 470k \cdot I_B + 0,7 + I_E \cdot 1k$$

$$I_B = \frac{12 - 0,7}{470k + (\beta + 1) \cdot 1k} = 18,19 \mu A \rightarrow (1,81 \times 10^{-5})$$

$$I_E = (\beta + 1) I_B \Rightarrow 151 \cdot 1,81 \times 10^{-5} = 2,75 mA$$

$$r_e = \frac{26mV}{I_{E_{DC}}} = \frac{26 \times 10^{-3}}{2,75 \times 10^{-3}} = \underline{\underline{9,45}}$$

$$\beta r_e = 150 \times 9,45 = \underline{\underline{1,418 k\Omega}}$$

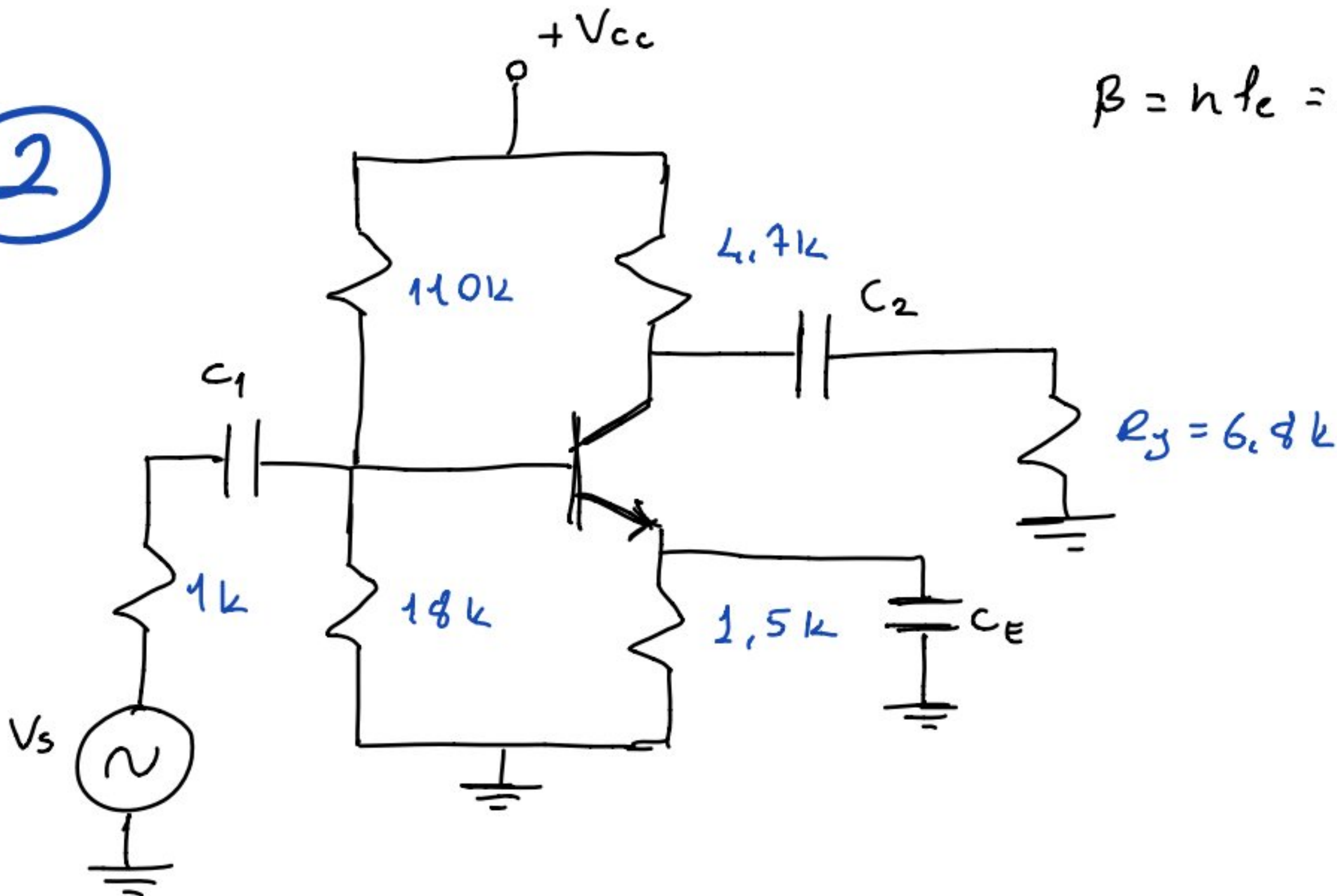
$$b) Z_i = 470k / \beta_{rc} \Rightarrow \frac{470k \cdot 1,414k}{470k + 1,414k} = \underline{\underline{1,413k \Omega}}$$

$$c) \underline{\underline{Z_o = 2,2k}}$$

$$d) A_v = \frac{V_o}{V_i} = \frac{-150i_b (2,2k // 3,3k)}{i_b \cdot 1,418k} = \frac{-198000i_b}{1,418ki_b} = \underline{\underline{-140}}$$

$$e) A_i = -A_v \frac{Z_i}{Z_y} = 140 \cdot \frac{1,413k}{3,3k} = \underline{\underline{60}}$$

2



$$\beta = h_{fe} = 200, h_{ie} = 1,2k$$

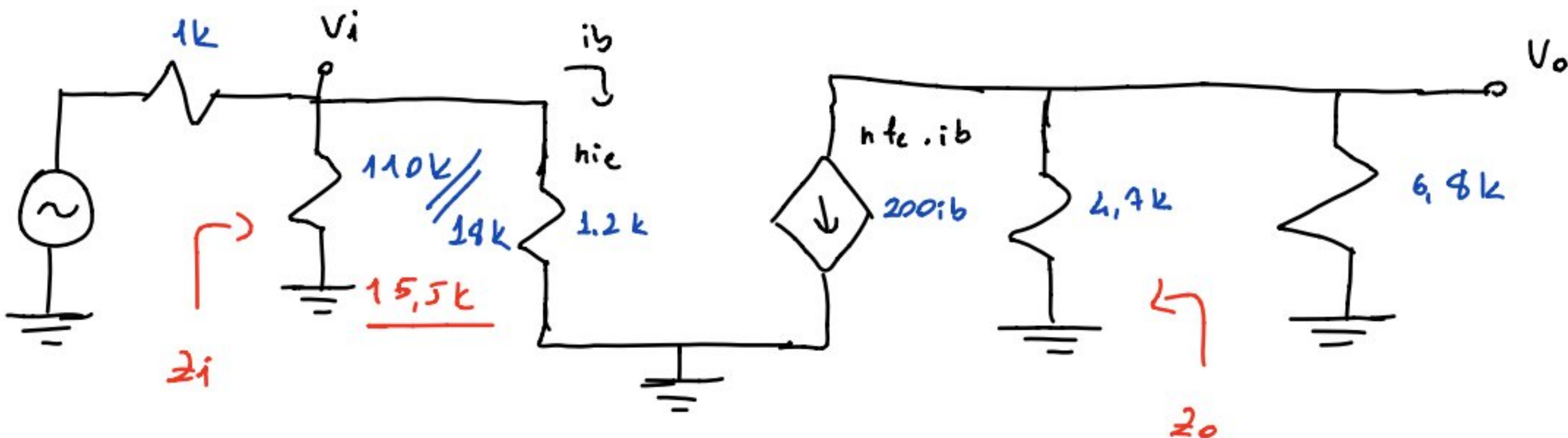
$$a) Z_i = ?$$

$$b) Z_o = ?$$

$$c) A_v = \frac{V_o}{V_i} = ?$$

$$d) A_{v_s} = \frac{V_o}{V_s} = ?$$

$$e) = \frac{i_o}{i_i} = ?$$



$$a) \quad Z_i = 15,5k // 1,2k = \underline{\underline{1,11k}}$$

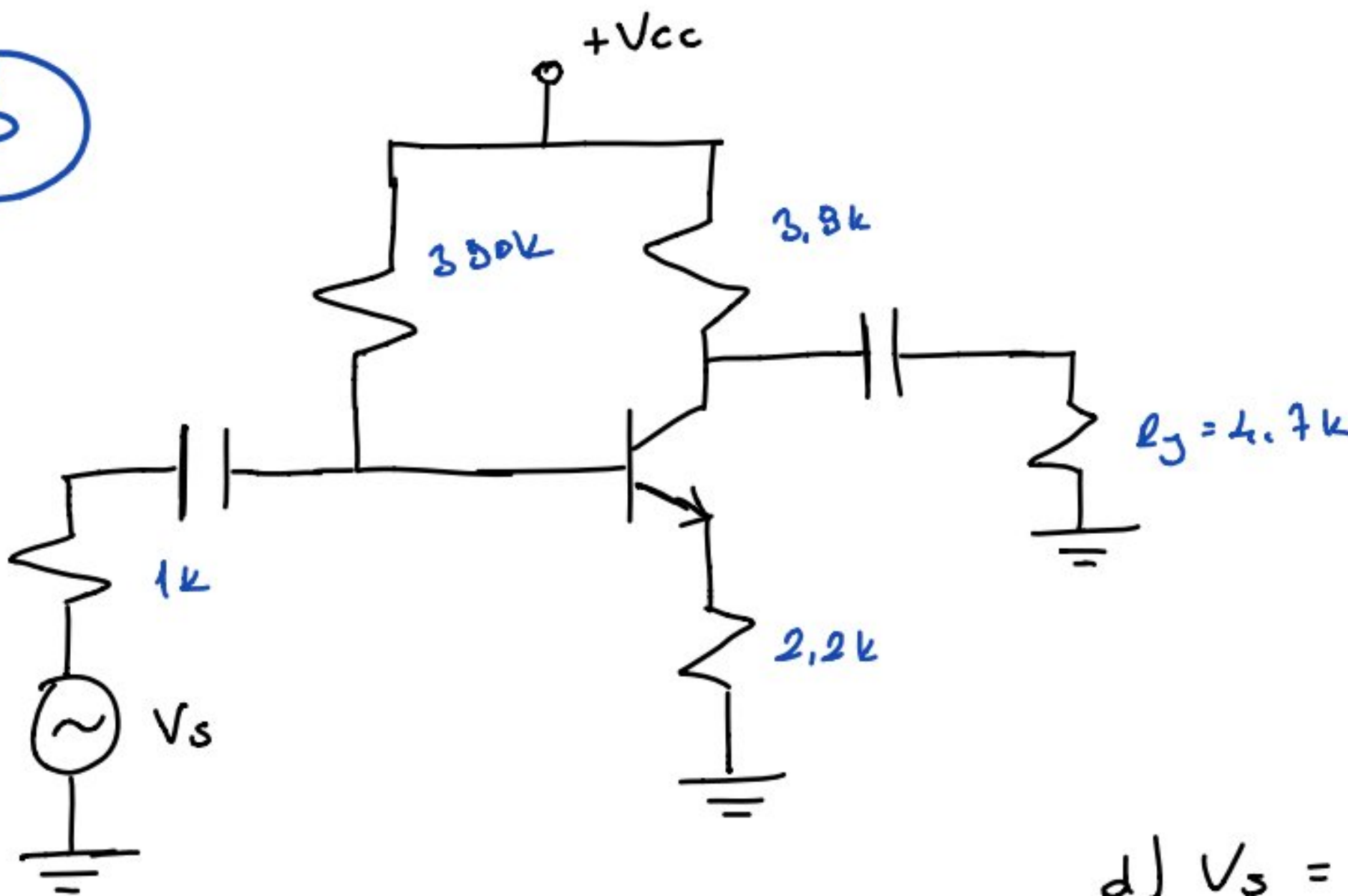
$$b) \quad Z_o = \underline{\underline{4,7k}}$$

$$c) \quad A_v = \frac{V_o}{V_i} = \frac{-200i_b (4,7k // 6,4k)}{1,2k i_b} = \underline{\underline{-463,2}}$$

$$d) \quad A_{v_s} = A_v \frac{Z_i}{Z_i + R_s} = -463,2 \frac{1,11k}{1,11k + 1k} = \underline{\underline{-243,67}}$$

$$e) \quad A_i = -A_v \frac{Z_i}{R_j} = 46,32 \frac{1,11k}{6,4k} = \underline{\underline{75,61}}$$

3



$$\beta_{ac} = 150 = h_{fe}$$

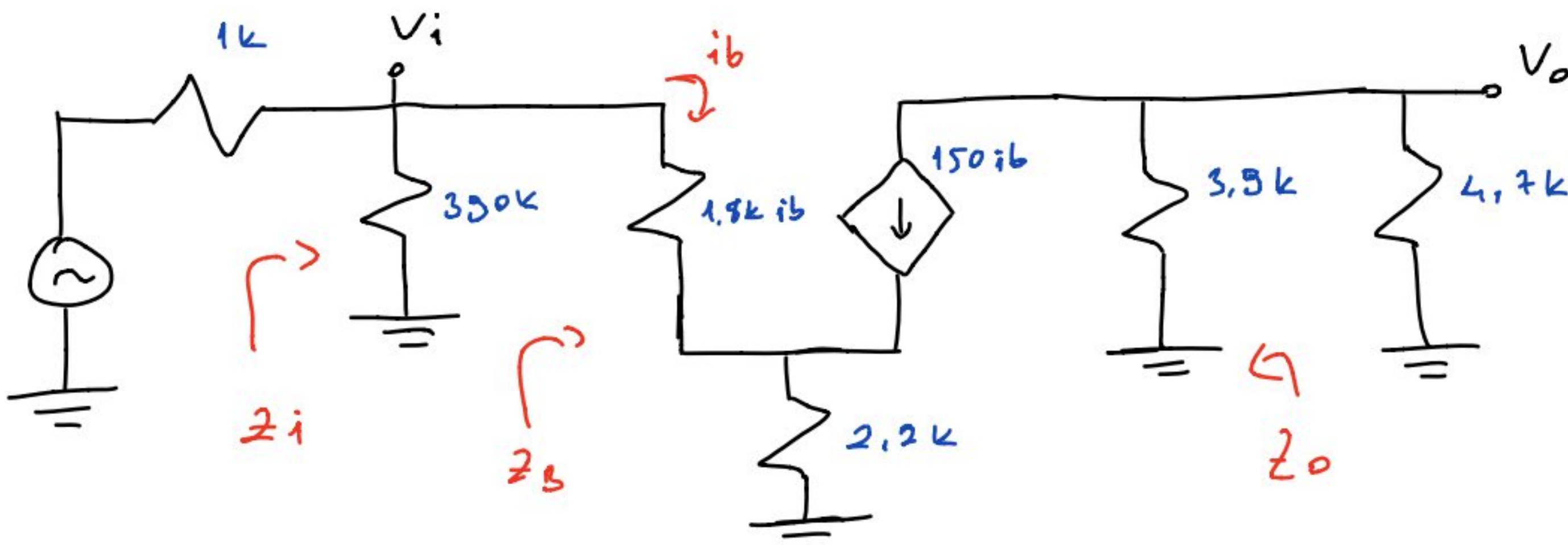
$$r_e = 12\Omega \rightarrow h_{ie} = 150 \cdot 12 = 1,8k$$

$$a) \quad Z_i, Z_o = ?$$

$$b) \quad A_v = \frac{V_o}{V_i} = ?$$

$$c) \quad A_{v_s} = \frac{V_o}{V_s} = ?$$

$$d) \quad V_s = 1mV \text{ sinus } i_{se} \quad V_o = ? \quad \text{cizimiz.}$$



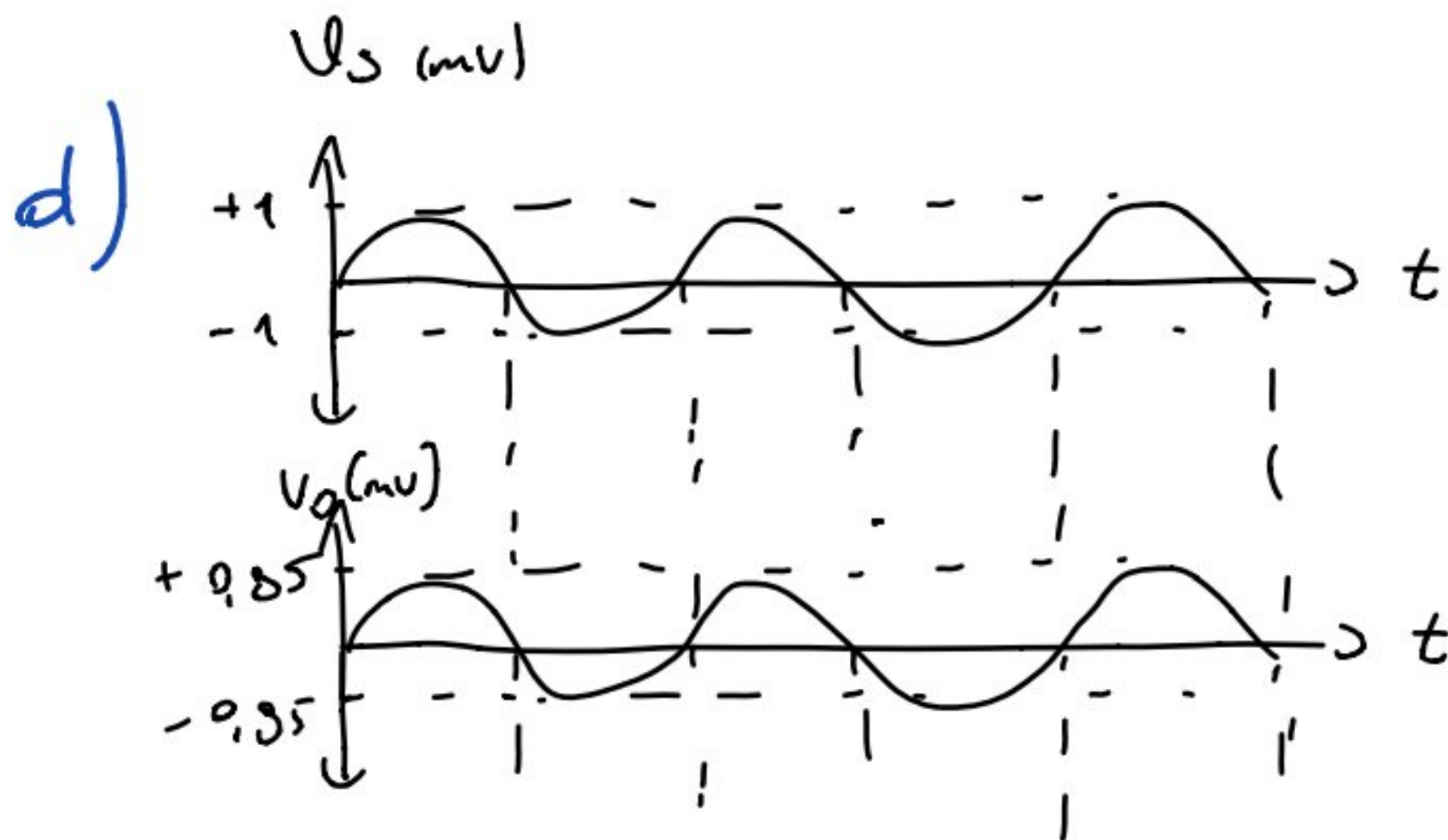
$$a) Z_i = 330k // Z_B \quad Z_B = \frac{V_i}{i_b} = \frac{1,8k \cdot i_b + 2,2k \cdot 151 i_b}{i_b} = \underline{\underline{334k}}$$

$$Z_i = 330k // 334k \approx 160k^*$$

$$Z_o = R_L = \underline{\underline{3,3k}}$$

$$b) A_v = \frac{V_o}{V_i} = \frac{-150 i_b (3,3k // 4,7k)}{1,8k \cdot i_b + 2,2k \cdot 151 i_b} = \frac{-320k i_b}{334k i_b} = \underline{\underline{-0,957}}$$

$$c) A_{v_s} = A_v \frac{Z_i}{Z_i + R_s} = -0,957 \frac{160k}{160k + 1k} = \underline{\underline{-0,951}}$$

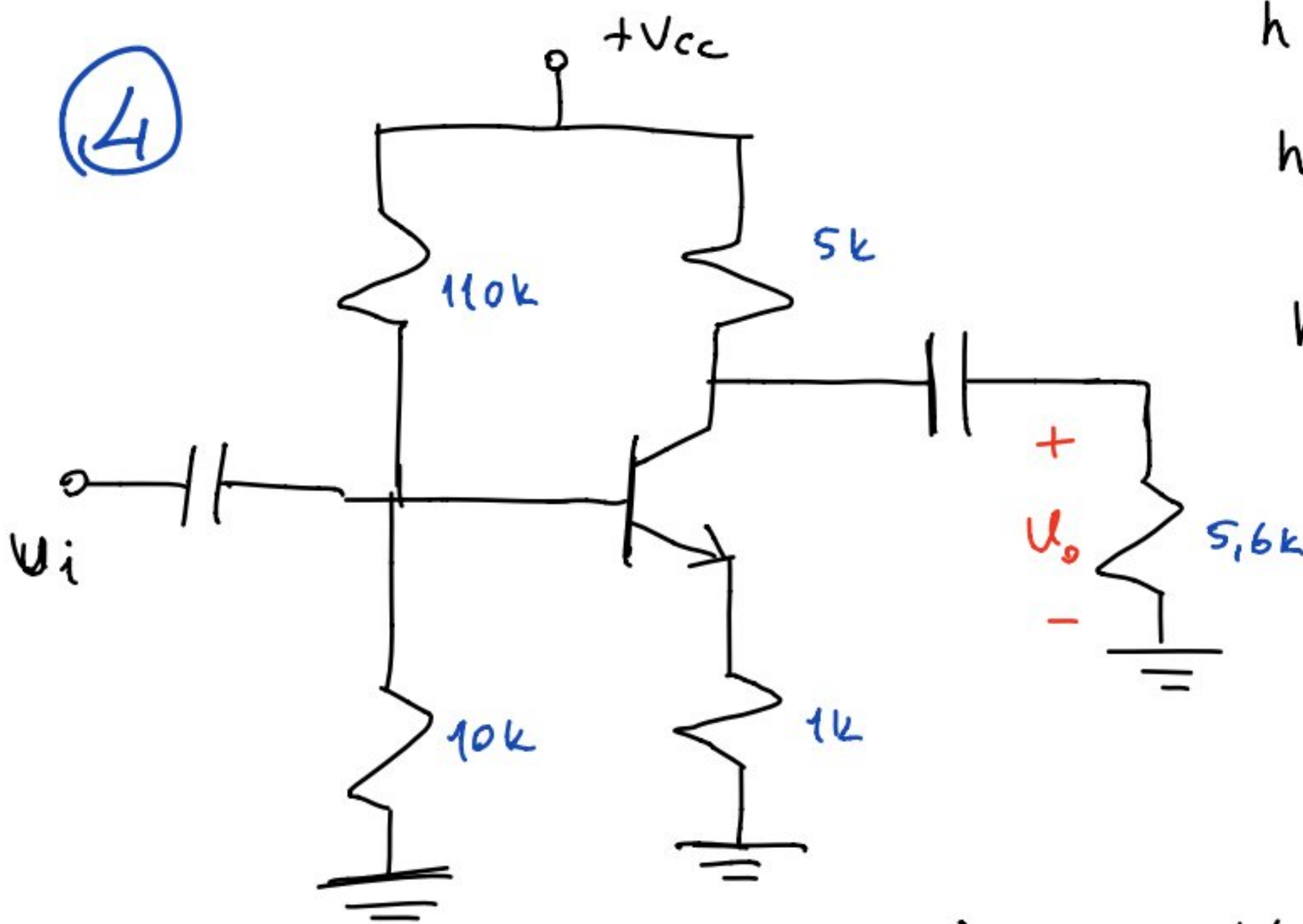


$$A_{v_s} = \frac{V_o}{V_s} =$$

$$-0,951 = \frac{V_o}{1mV}$$

$$V_o = -0,95mV$$

④



$$h_{fe} = 110 = \beta$$

$$h_{ie} = 1.1k = \beta r_e$$

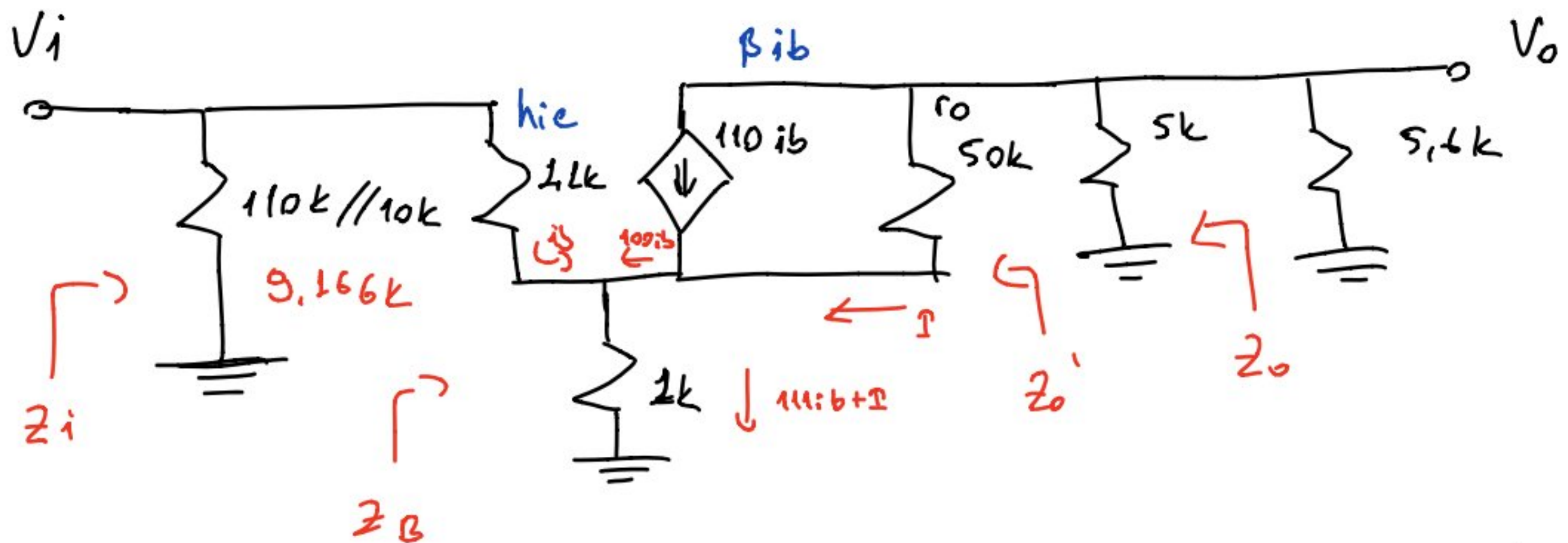
$$h_{oe} = 2\mu S$$

$$r_o = \frac{1}{h_{oe}} = 50k$$

a) $Z_i = ?$

b) $Z_o = ?$

c) $A_v = \frac{V_o}{V_i} = ?$ d) $A_i = \frac{i_o}{i_i} = ?$



a) $Z_i = 9.166k // Z_B$ $Z_B = \frac{V_i}{i_b} = \frac{1.1k \cdot i_b + 1k(111i_b + I)}{i_b}$

$I = ?$

Çevre $5k // 5.6k (110i_b + I) + 50k \cdot I + 1k(111i_b + I) = 0$

$$230.5k i_b + 2.64k \cdot I + 50k \cdot I + 111k i_b + 1k I = 0$$

$$402.56k i_b + 53.64k I = 0$$

$$I = - \frac{402.56k i_b}{53.64k} = -7.486 i_b$$

$$Z_8 = \frac{1,1k \cancel{ib} + 111k \cancel{ib} - 7,486k \cancel{ib}}{\cancel{ib}} = \underline{\underline{104,614k \Omega}}$$

$$Z_i = 3,166k // 104,614k = \underline{\underline{8,428k \Omega}}$$

b) $Z_o = 5k // Z_o'$

$$Z_o' = \frac{V_x}{110ib + I} = \frac{50k \cdot I + 1k(111ib + I)}{110ib + I}$$

Çevre

$$3,766kib + 1,1kib + 1k(111ib + I) = 0$$

$$3,766kib + 1,1kib + 111kib + 1k \cdot I = 0$$

$$I = \frac{-121,866kib}{1k} = \underline{\underline{-121,86ib}}$$

$$Z_o' = \frac{50k(-121,86ib) + 1k(111ib - 121,86ib)}{110ib - 121,86ib}$$

$$Z_o' = \frac{-6093000ib + 111kib - 121,86kib}{-11,86ib}$$

$$Z_o' = 514,653k \Omega$$

$$Z_o = 5k // 514,653k \Omega$$

$$= \underline{\underline{4,851k \Omega \approx 5k \Omega}}$$

c)

$$A_v = \frac{V_o}{V_i} \Rightarrow \frac{- \overbrace{(5k // 5.6k)}^{2,641k} \overbrace{(110ib + I)}^{102,514ib}}{111ib + 1k \underbrace{(111ib + I)}_{103,514ib}}$$

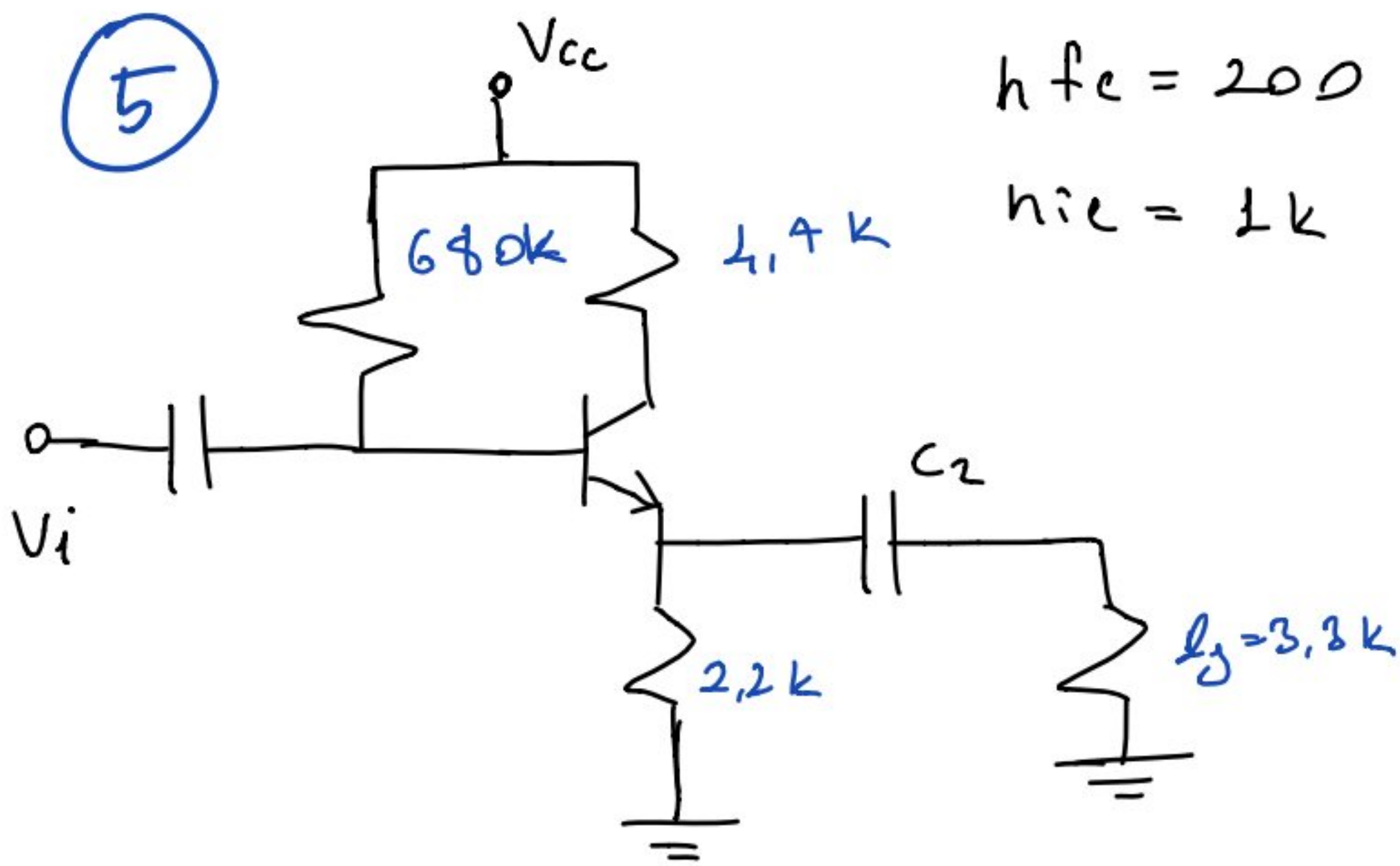
$$I = \underline{\underline{-7,486ib}}$$

$$A_v = \frac{-270739,4ib}{104614ib} = \underline{\underline{-2,5879}}$$

d)

$$A_i = -A_v \frac{Z_i}{R_j} = -2,5879 \frac{8,428k}{5,6k}$$

$$\underline{\underline{A_i = -3,8347}}$$



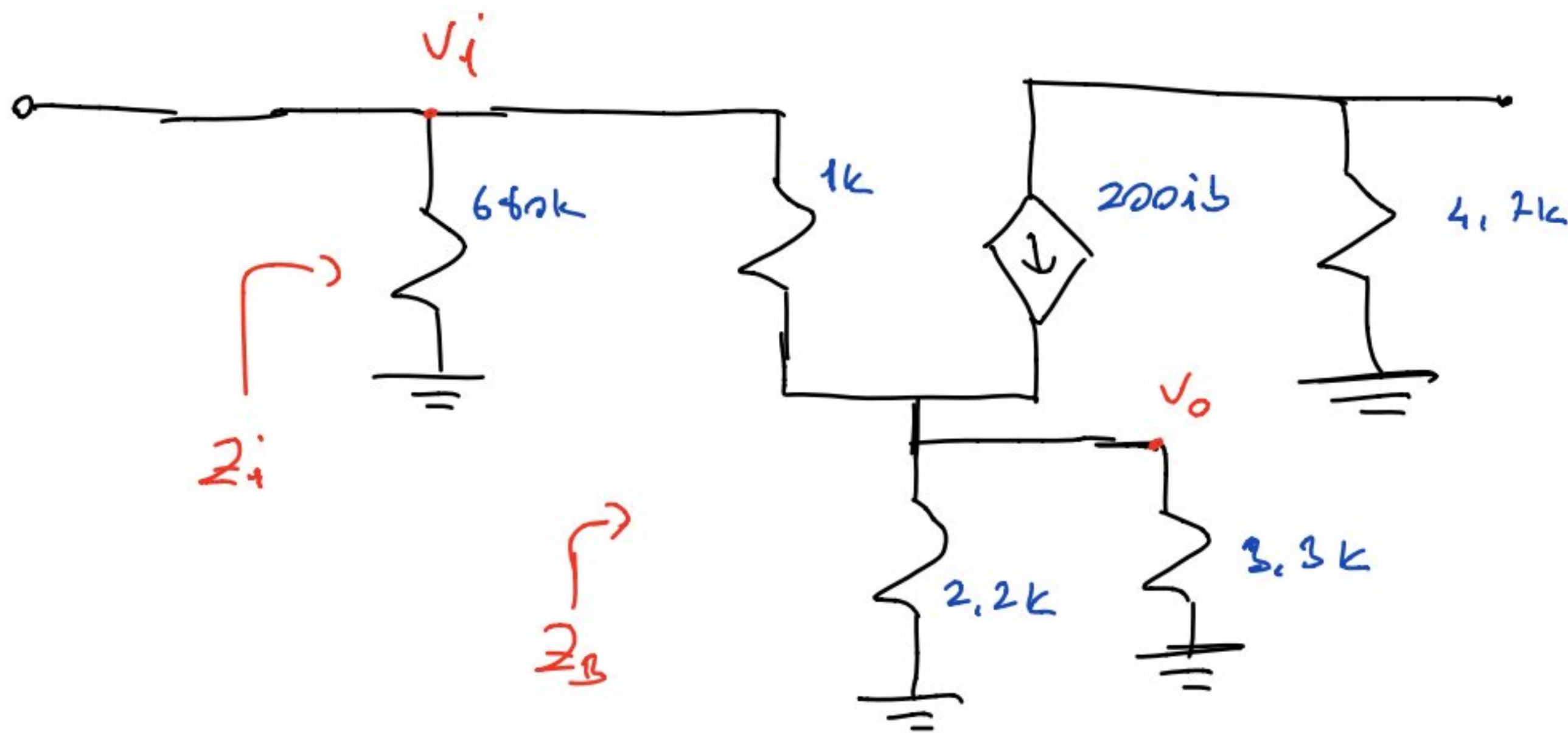
a) $Z_i = ?$

b) $Z_o = ?$

c) $A_u = \frac{V_o}{V_i} = ?$

d) $A_i = \frac{i_o}{i_i} = ?$

e) girişe kaynak iç direnci $1k$ olan bir V_s bağlarsa $Z_o = ?$



a) $Z_i = 680k // Z_B$

$$Z_B = \frac{V_i}{i_b} = \frac{i_b \cdot 1k + 200i_b (2.2k // 3.3k)}{i_b}$$

$$= \underline{266,3k}$$

$$Z_i = 680k // 266,3k$$

$$= \underline{191,36k}$$

$$Z_o = R_E // \frac{h_{ie}}{1 + h_{fe}} = 2,2k // \frac{1k}{1 + 200} = \underline{4,858\Omega}$$

b) (U_i varken)

$$Z_o = \frac{V_T}{I_T} = \frac{(201 \mu A + I_T) \cdot 2.2 \text{ k}}{I_T}$$

$$680 \text{ k} \cdot i_b + 1 \text{ k} \cdot i_b + 2.2 \text{ k} (201 \mu A + I_T) = 0$$

$$680 \text{ k} \cdot i_b + 1 \text{ k} \cdot i_b + 442.2 \text{ k} \cdot i_b + 2.2 \text{ k} \cdot I_T = 0$$

$$I_T = -510.54 \mu A$$

$$Z_o = \frac{(201 \mu A - 510.54 \mu A) \cdot 2.2 \text{ k}}{-510.54 \mu A} = \underline{\underline{1333.85 \Omega}}$$

$$c) A_v = \frac{U_o}{U_i} = \frac{-201 \mu A (2.2 \text{ k} / 13.3 \text{ k})}{\underbrace{28 \cdot i_b}_{266.3 \text{ k} \mu A}} = \underline{\underline{-0.886}}$$

$$d) A_i = -A_v \frac{Z_i}{Z_j} = 0.886 \frac{181.36 \text{ k}}{3.8 \text{ k}} = \underline{\underline{57.75}}$$

e) (U_s varoa formül)

$$Z_o = R_E // \frac{(R_o // R_g) + h_{ie}}{1 + h_{fe}} = 2.2 \text{ k} // \frac{(1 \text{ k} // 680 \text{ k}) + 1 \text{ k}}{1 + 200} = \underline{\underline{9.90 \Omega}}$$

$$Z_o = \frac{V_T}{I_T} = \frac{(201 \mu A + I_T) \cdot 2.2 \text{ k}}{I_T}$$

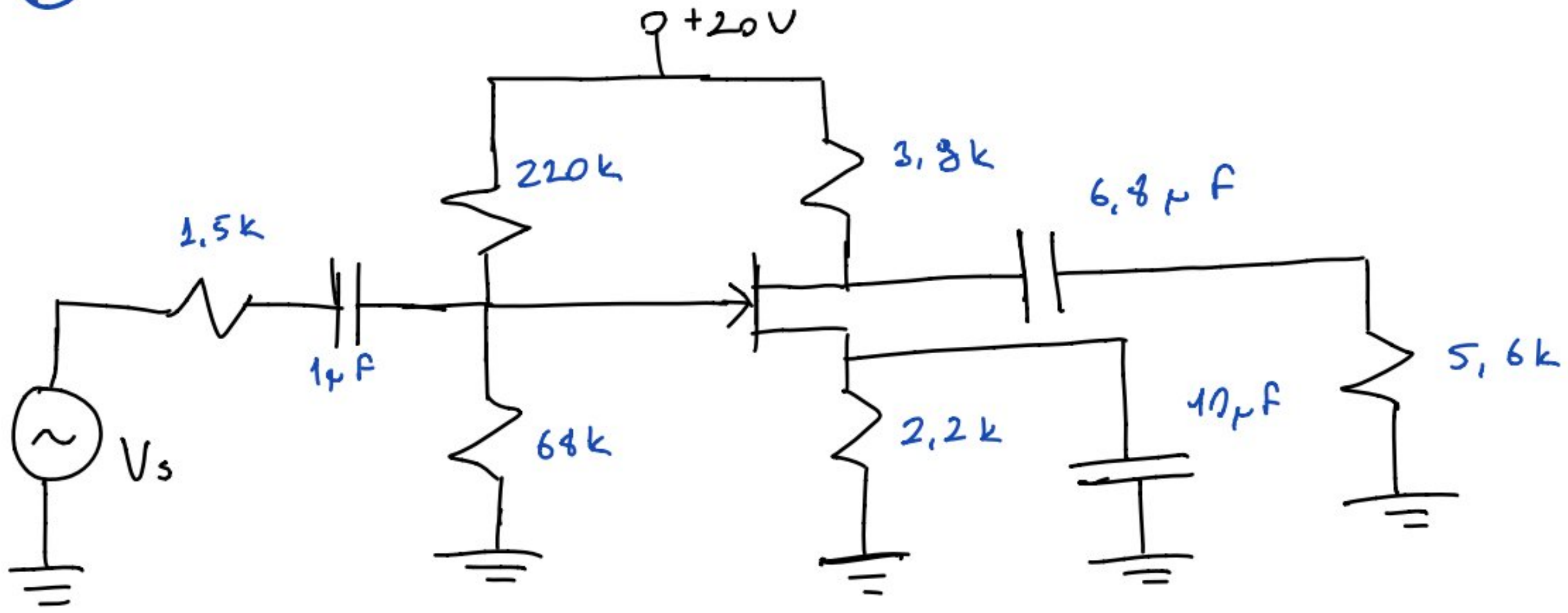
$$(1 \text{ k} // 680 \text{ k}) \cdot i_b + 1 \text{ k} \cdot i_b + 2.2 \text{ k} (201 \mu A + I_T) = 0$$

$$338.53 \mu A + 1 \text{ k} \cdot i_b + 442.2 \text{ k} \cdot i_b + 2.2 \text{ k} \cdot I_T = 0$$

$$I_T = -201.90 \mu A$$

$$Z_o = \frac{(201 \mu A + 201.9 \mu A) \cdot 2.2 \text{ k}}{-201.90 \mu A} = \underline{\underline{9.833 \Omega}}$$

6



$$I_{DSS} = 10\text{mA} \quad V_p = -6\text{V} \quad C_{wi} = 4\text{pF} \quad C_{wo} = 6\text{pF}$$

$$C_{gd} = 8\text{pF} \quad C_{gs} = 12\text{pF}$$

$$C_{ds} = 3\text{pF}$$

"Üstteki devre için;

a) DC analiz yaparak g_m değerini bulunuz.

b) AC eşdeğerini çiziniz

c) Z_i , Z_o , A_v , A_{vs} değerini hesaplayınız.

a) $I_G = 0\text{A}$ olduğu için

$$V_G = \frac{20\text{V}}{(220\text{k} + 64\text{k})} \cdot 64\text{k} = \underline{\underline{4,72\text{V}}} \quad \checkmark$$

$$V_G = V_{GS} + V_{RS} \Rightarrow 4,72\text{V} = V_{GS} \cdot I_S \cdot (2,2\text{k})$$

$$V_{GS} = 4,72\text{V} - I_D (2,2\text{k}) \quad \checkmark$$

$$I_D = \frac{4,72 - V_{GS}}{2,2k}$$

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2 \Rightarrow$$

$$\frac{4,72 - V_{GS}}{2,2k} = 10mA \left(1 - \frac{V_{GS}}{-6} \right)^2$$

$$36 / \frac{4,72 - V_{GS}}{22} = 1 + \frac{V_{GS}}{3} + \frac{V_{GS}^2}{36}$$

$$4,72 - 1,63V_{GS} = 36 + 12V_{GS} + V_{GS}^2$$

$$V_{GS}^2 + 13,63V_{GS} + 29,28 = 0$$

$$\sqrt{b^2 - 4ac} = \underline{\underline{8,52}}$$

$$x_1 = \frac{-b - \Delta}{2a} = -11,075$$

$$x_2 = \frac{-b + \Delta}{2a} = -2,555 = V_{GS}$$

$$g_m = \frac{2 I_{DSS}}{|V_p|} \left(1 - \frac{V_{as}}{V_p} \right)$$

$$g_m = \left| \frac{2 \cdot 10 \cdot 10^{-3}}{-6} \right| \left(1 - \frac{-2,555}{-6} \right)$$

$$g_m = 3,333 \cdot 0,5744$$

$$g_m = 1,9132$$